IGCSE 0607 Practice Questions — E6.3

Circles, arcs and sectors

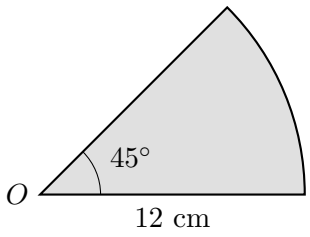
Total Questions: 18 — **Total Marks:** 130 Marks

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- **E6.3** Circles, arcs and sectors
  → **Paper 2 (Non-Calculator)** *(8 Qs)* · **Paper 4 (Calculator)** *(10 Qs)*

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1 The diagram shows a sector of a circle, center O, with radius 12 cm and sector angle 45°.



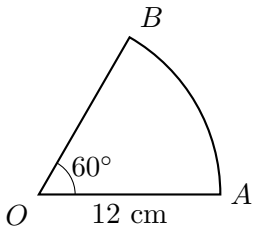
(a) Work out the exact length of the arc of the sector.
Give your answer in terms of π.

..... cm **[2]**

(b) Work out the exact area of the sector.
Give your answer in terms of π.

..... cm^2 **[2]**

2 The diagram shows a minor sector AOB of a circle with centre O and radius 12 cm.
The sector angle is 60°.



NOT TO SCALE

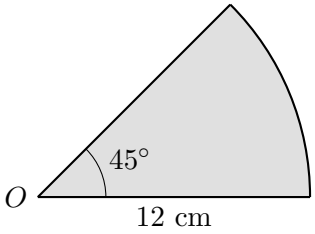
(a) Work out the length of the minor arc AB.
Give your answer in terms of π.

..... cm **[2]**

(b) Work out the area of the minor sector AOB.
Give your answer in terms of π.

..... cm^2 **[2]**

3 The diagram shows a sector of a circle with centre O and radius 12 cm.
The sector angle is 45°.



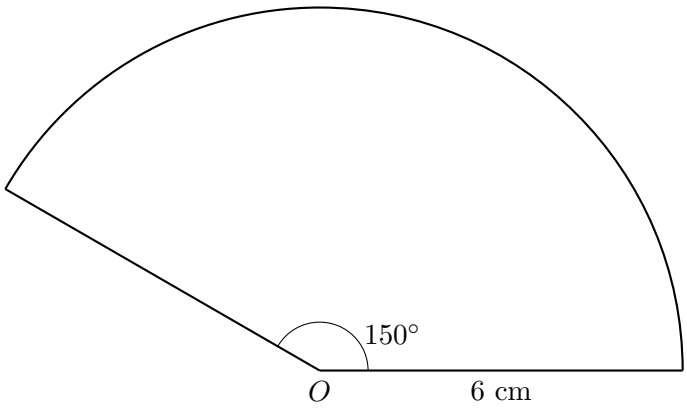
(a) Work out the exact area of the sector.
Give your answer in terms of π.

Area = ... cm^2 **[2]**

(b) Work out the exact perimeter of the sector.
Give your answer in terms of π.

Perimeter = ... cm **[3]**

4 The diagram shows a sector of a circle with centre O and radius 6 cm.
The sector angle is 150°.



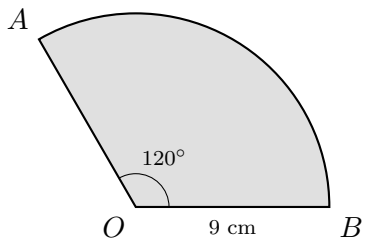
(a) Calculate the area of the sector.
Give your answer in terms of π.

..... cm^2 **[2]**

(b) Calculate the perimeter of the sector.
Give your answer in terms of π.

..... cm **[3]**

5 The diagram shows a sector OAB of a circle with centre O and radius 9 cm.
The sector angle $\angle AOB = 120°$.



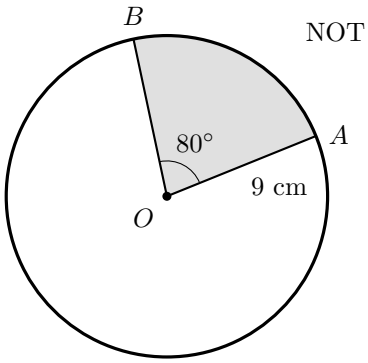
(a) Calculate the length of the minor arc AB.
Give your answer in terms of π.

..... cm **[2]**

(b) Calculate the area of the shaded minor sector OAB.
Give your answer in terms of π.

Area = ... cm^2 **[3]**

6 The diagram shows a circle, centre O, of radius 9 cm. Points A and B lie on the circumference such that the minor sector angle $AOB = 80°$.



(a) Giving your answers in terms of π, find

(i) the length of the minor arc AB,

..... cm **[2]**

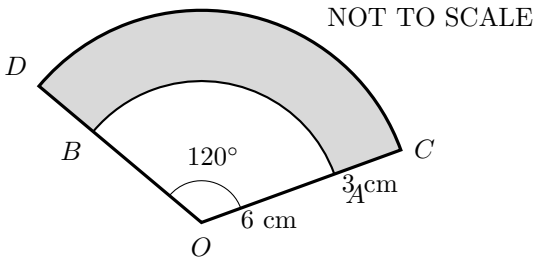
(ii) the perimeter of the minor sector OAB.

..... cm **[1]**

(b) Calculate the area of the major sector OAB, giving your answer in terms of π.

..... cm^2 **[3]**

7 The diagram shows two sectors OAB and OCD with common centre O and sector angle 120°.
$OA = OB = 6$ cm and $AC = BD = 3$ cm. The region $ABDC$ is shaded.



(a) Calculate

(i) the length of arc AB, giving your answer in terms of π,

..... cm **[2]**

(ii) the perimeter of the shaded region $ABDC$, giving your answer in the form $p + q\pi$, where p and q are integers.

..... cm **[3]**

(b) Calculate the area of the shaded region $ABDC$, giving your answer in terms of π.

..... cm^2 **[3]**

(c) A different sector OPQ has radius r cm and sector angle 150°.
The area of sector OPQ is equal to the area of the shaded region $ABDC$.
Calculate the value of r.

$r =$ ..... **[2]**

8 **(a)** A sector of a circle, center O, has radius r cm and sector angle $\theta°$.
The area of the sector is 10π cm^2 and the length of its arc is 2π cm.

Find the value of r and the value of θ.

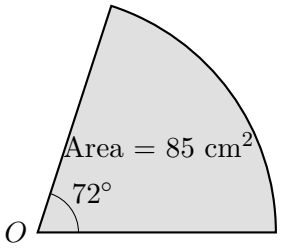
$r =$ ..... , $\theta =$ ..... **[5]**

(b) Another sector of a circle has radius x cm and sector angle $(x + 20)°$.
The length of its arc is $\frac{16\pi}{5}$ cm.

Find the value of x.
You must show all your working.

$x =$ ..... **[4]**

9 The diagram shows a sector of a circle with centre O and angle 72°.
The area of the sector is 85 cm^2.



NOT TO SCALE

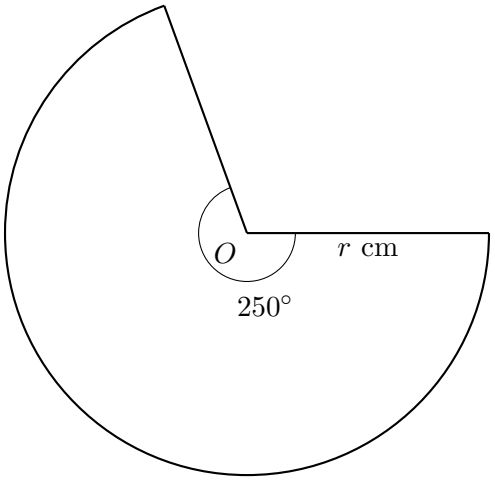
(a) Calculate the radius of the sector.

..... cm **[3]**

(b) Calculate the perimeter of the sector.

..... cm **[3]**

10 The diagram shows a major sector of a circle with centre O and radius r cm.
The reflex angle is 250°.
The length of the major arc is 35 cm.



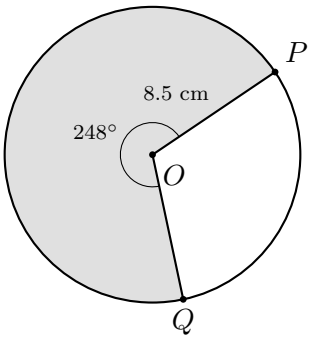
(a) Calculate the value of r.

$r =$ ..... **[3]**

(b) Calculate the area of the major sector.

..... cm^2 **[3]**

11 The diagram shows a circle, centre O, with radius 8.5 cm.
Points P and Q lie on the circumference such that the reflex angle $POQ = 248°$.



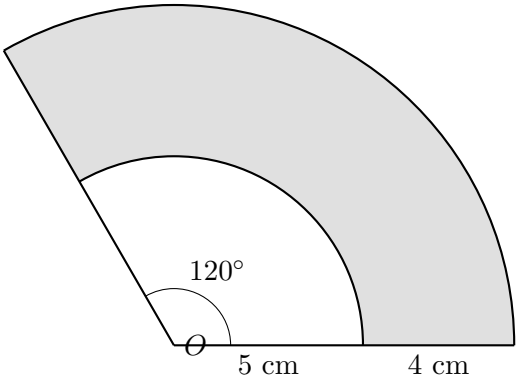
(a) Calculate the perimeter of the shaded major sector OPQ.

Perimeter = ... cm **[3]**

(b) Calculate the area of the unshaded minor segment bounded by the chord PQ and the minor arc PQ.

Area = ... cm^2 **[3]**

12 The diagram shows a shaded shape made from two concentric sectors of circles with centre O.
The inner radius is 5 cm and the outer radius is 9 cm.
The angle of both sectors is 120°.



(a) Calculate the area of the shaded shape.

Area = ... cm^2 **[3]**

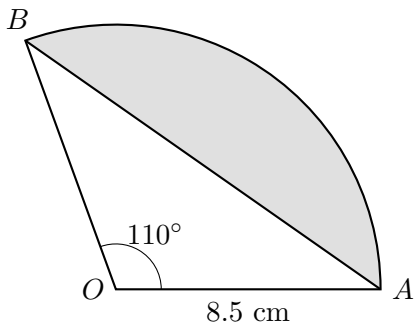
(b) Calculate the perimeter of the shaded shape.

Perimeter = ... cm **[3]**

(c) A complete circle has the same area as the shaded shape.
Calculate the radius of this circle.

Radius = ... cm **[2]**

13 The diagram shows a sector OAB of a circle with center O and radius 8.5 cm.
The sector angle is 110°.
A straight line connects A and B, forming a triangle OAB and a shaded segment.



(a) Calculate the area of the sector OAB.

..... cm^2 **[2]**

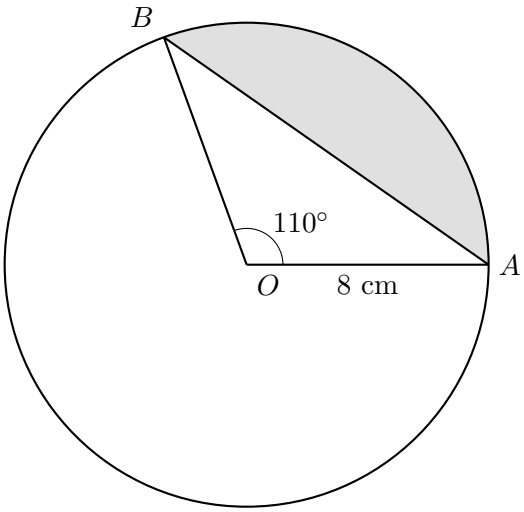
(b) Calculate the area of the shaded segment.

..... cm^2 **[3]**

(c) Calculate the perimeter of the shaded segment.

..... cm **[4]**

14 The diagram shows a circle with centre O and radius 8 cm.
A and B are points on the circumference such that angle $AOB = 110°$.



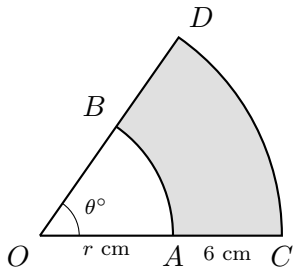
(a) Calculate the area of the shaded segment.

..... cm^2 **[4]**

(b) Calculate the perimeter of the shaded segment.

..... cm **[3]**

15 The diagram shows two concentric sectors OAB and OCD with centre O and sector angle $\theta°$.
$OA = r$ cm and $AC = 6$ cm.
The area of the shaded region $ABDC$ is 54π cm^2.



(a) Show that $\theta = \dfrac{1620}{r+3}$.

[4]

(b) The total perimeter of the shaded region $ABDC$ is 54 cm.

(i) Calculate the value of r.

$r =$ ..... **[3]**

(ii) Find the value of θ.

$\theta =$ ..... **[1]**

16 A sector of a circle has radius r cm and sector angle $x°$.

(a) The arc length of the sector is 6π cm and the area of the sector is 27π cm^2.

(i) Work out the value of r.

$r =$ ..... **[3]**

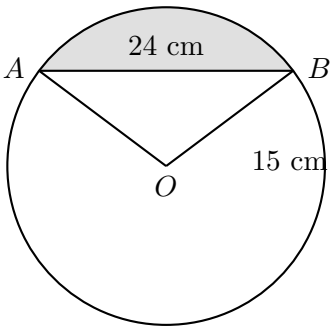
(ii) Work out the value of x.

$x =$ ..... **[2]**

(b) Another sector has radius R cm and sector angle $y°$.
The perimeter of this sector is equal to the circumference of a circle with radius R cm.
Find an expression for y in terms of π.

$y =$ ..... **[4]**

17 The diagram shows a circle, centre O, with radius 15 cm.
AB is a chord of the circle with length 24 cm.
The region bounded by the chord AB and the minor arc AB is shaded.



NOT TO SCALE

(a) Show that angle $AOB = 106.3°$, correct to 1 decimal place.

[3]

(b) For the following calculations, use angle $AOB = 106.3°$.

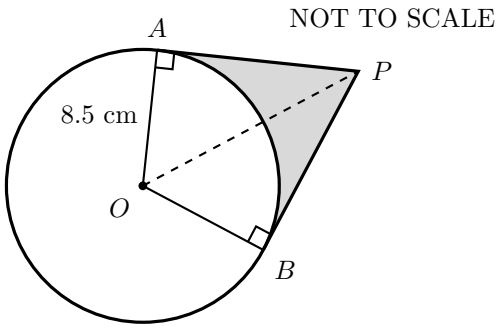
(i) Calculate the perimeter of the shaded minor segment.

..... cm **[3]**

(ii) Calculate the area of the shaded minor segment.

..... cm^2 **[4]**

18 The diagram shows a circle, centre O, of radius 8.5 cm. From an external point P, tangents PA and PB are drawn to touch the circle at A and B respectively, where $OP = 15.2$ cm. The region bounded by PA, PB, and the minor arc AB is shaded.



(a) Show that obtuse angle $AOB = 112.0°$, correct to 1 decimal place.

[3]

(b) Calculate the length of the major arc AB, giving your answer correct to 3 significant figures.

..... cm **[3]**

(c) Calculate the area of the shaded region, giving your answer correct to 3 significant figures.

..... cm^2 **[4]**

(d) A sector of a different circle has radius R cm, arc length 18.4 cm, and area 115 cm^2. Calculate the value of R and the sector angle θ, giving θ correct to 1 decimal place.

$R =$ ..... , $\theta =$ ..... **[3]**